Memory Error While Running Quantum SVM (QSVM)

While experimenting with the quantum version of the SVM classifier using Qiskit Machine Learning's **QuantumKernel**, I encountered a significant challenge related to memory usage.

After successfully training and evaluating the classical SVM model with reasonable performance:

Accuracy: 0.7307
Precision: 0.7471
Recall: 0.6989
F1-score: 0.7222

I proceeded to run the quantum SVM model on the same dataset. However, during the **fit()** step of the **QSVM**, the following critical error was encountered:

numpy._core._exceptions._ArrayMemoryError: Unable to allocate 23.4 GiB for an array with shape (56000, 56000) and data type float64

This error was triggered by the **QuantumKernel.evaluate()** method internally used by Qiskit's QSVM implementation. The cause stems from the exponential scaling of the kernel matrix with respect to the number of data points. Specifically, the kernel matrix computation attempts to generate a full pairwise similarity matrix of shape (n_samples, n_samples)—which in my case (likely due to a larger dataset split or parameter setting) resulted in an attempt to allocate 23.4 GB of RAM, far exceeding typical system memory availability.

Key Takeaways:
- **Root Cause:** The quantum kernel computes a similarity matrix between every pair of samples in X_train, leading to memory explosion with even moderately sized datasets.
- **Trigger Point:** model.fit(X_train, y_train) where model is an instance of QSVC(kernel=quantum_kernel).
- **Implication:** Quantum kernels are currently best suited for smaller datasets, especially in simulation environments where classical memory constraints still apply.
- **Resolution Direction:** To proceed, I plan to:
• Reduce the training set size (e.g., via stratified sampling or dimensionality reduction).
• Switch to approximate kernel evaluation strategies (if available).
• Explore running on actual quantum hardware where kernel evaluations are not stored explicitly in large arrays.

Reflection:
This incident highlighted a core limitation of current hybrid quantum-classical machine learning approaches in practical settings. While classical models scale well with more data, quantum-enhanced models are still bound by simulation overheads and memory bottlenecks. This experience was an eye-opener and reaffirms the need for smart dataset preprocessing and careful consideration when applying quantum algorithms to real-world problems.